Basics of FE-Analysis BK10A6400 Lecture 6: Numerical integration, simple examples

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Numerical integration

- Basis of numerical integration (MathWorld), for example.
- To find an approximate solution to a definite integral $\int_{-1}^{1} f(x)dx \approx \sum_{i=1}^{nip} f(x_i)w_i$.
- Needed to evaluate area or volume integral to get $\overline{\boldsymbol{K}}$
- Analytically before - Why make it numerically?
  - Faster (for complicated element)
  - Analytical solution is not always possible
  - Specific numerical integration schemes are needed to get rid of locking phenomena
- Gaussian quadratures are used because a polynomial of degree $p = 2nip - 1$ can be integrated exactly with nip integration points

$$W_{int} = \frac{1}{2}\int_V \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\mathrm{d}V = \frac{1}{2}\boldsymbol{u}^T \underbrace{\int_V \boldsymbol{B}^T \boldsymbol{D}\boldsymbol{B}\,\mathrm{d}V}_{\overline{\boldsymbol{K}}}\,\boldsymbol{u} \qquad (1)$$

$$W_{int} = \frac{1}{2}l_z \int_A \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\mathrm{d}A = \frac{1}{2}\boldsymbol{u}^T \underbrace{l_z \int_A \boldsymbol{B}^T \boldsymbol{D}\boldsymbol{B}\,\mathrm{d}A}_{\overline{\boldsymbol{K}}}\,\boldsymbol{u} \qquad (2)$$

Use Gauss quadrature integration scheme to estimate function $A(x) = \int_a^b f(x)dx$ where $f(x) = x^2$ and $a = 0, b = 3$.

$$A = \int_a^b f(x)dx = \int_0^b x^2 dx \tag{3}$$

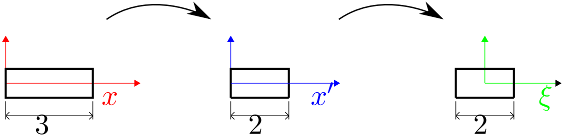
Transformation: $x \rightarrow \xi$

$$x = \frac{b}{2}(\xi + 1) \quad ; dx = \frac{b}{2} d\xi \tag{4}$$

$$A = \int_{-1}^{1} f(\xi)\frac{b}{2} d\xi \approx \sum_{i=1}^{nip} f(\xi_i) w_i \frac{b}{2} \tag{5}$$

Matlab class assigment: transformation $x \rightarrow \xi$ step by step

Scale $f : x \rightarrow x'$ Translate $f : x' \rightarrow \xi$



Scaling $x \rightarrow x'$ (from coordinates 0 ... 3 $\rightarrow 0 \ldots 2)$:

$$x' = \frac{2}{3}x \quad ; \quad x'(x=0) = 0\,;\, x'(x=3) = 2 \tag{6}$$

Translating $x' \rightarrow \xi$ (from coordinates 0 ... 2 $\rightarrow -1 \ldots 1)$:

$$\xi = x' - 1 \quad ; \quad \xi(x'=0) = -1\,;\, \xi(x'=2) = 1 \tag{7}$$

Substituting x', final eq:

$$\xi = \frac{2}{3}x - 1 \quad ; \quad \xi(x=0) = -1\,;\, \xi(x=3) = 1 \tag{8}$$

Matlab class assigment: transformation $x \rightarrow \xi$ step by step

Now x can be solved from Eq. (8):

$$x = \frac{3}{2}(\xi + 1) \tag{9}$$

and differential dx:

$$\frac{dx}{d\xi} = \frac{3}{2} \quad \rightarrow \quad dx = \frac{3}{2} d\xi \tag{10}$$

Now the area $A(x)$ can be written

$$A = \int_0^3 x^2 dx = \int_{-1}^1 \underbrace{\left(\frac{3}{2}(\xi + 1)\right)^2}_{f(\xi)} \underbrace{\frac{3}{2} d\xi}_{dx} \tag{11}$$

Matlab class assigment: Numerical integration

A number of integration points for an accurate solution. The degree of a polynomial of the integrand $p = 2$.

$$nip = \frac{(p+1)}{2} = 1.5 \tag{12}$$

so two integration points are needed. For Gaussian quadrature integration, integration points and weights can be found in mathematics handbooks.

Matlab class assignment: Numerical integration in Matlab

Two point Gauss quadrature formula is exact for polynomials up to degree 2:

```
xiv=[-0.577350269189 63,0.57735026918963];
wxiv=[1,1];
```

You need to write function where is your integrand $f(x) = x^2$ and then main code where integration is done:

```
f=0;
for i=1:nip,
   f=f+wxiv(i)*fdxi(xiv(i));
end
```

Numerical integration in 2D and 3D

2D:

$$\int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta)\,\mathrm{d}\xi\mathrm{d}\eta \approx \sum_{j=1}^{nip_\eta}\sum_{i=1}^{nip_\xi} f(\xi_i,\eta_j) w_{\xi_i} w_{\eta_j} \tag{13}$$

3D:

$$\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta,\zeta)\,\mathrm{d}\xi\mathrm{d}\eta \approx \sum_{k=1}^{nip_\zeta}\sum_{j=1}^{nip_\eta}\sum_{i=1}^{nip_\xi} f(\xi_i,\eta_j,\zeta_j) w_{\xi_i} w_{\eta_j} w_{\zeta_k} \tag{14}$$

Shear locking and selective reduced integration

To get rid of shear locking (of bilinear plate element), you should make separated integration for energies related to shear and normal deformations.

$$U = U_n + U_s \tag{15}$$

$$U_n = \frac{1}{2} l_z \int_V \boldsymbol{\sigma}_n^T \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V \boldsymbol{\varepsilon}_n^T \boldsymbol{D}_n \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{16}$$

$$U_s = \frac{1}{2} l_z \int_V \sigma_{xy} \gamma_{xy} \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V D_s \gamma_{xy}^2 \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{17}$$